

Week 4 Assessment Exercise 4

Mariam Serhan

2026-10-05

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Introduction to the NHANES Data Set

The National Health and Nutrition Examination Survey (NHANES) is a program run by the CDC's National Center for Health Statistics (NCHS) and its purpose is to assess the health and nutrition of the American population. (Health Statistics 2026) Their data is released to the public and can be acquired using this link here: <https://wwwn.cdc.gov/nchs/nhanes/>

In this assignment, waves 2015, 2016 and 2017 will be used.

Importing Data and Loading Packages

```
library(pacman)

pacman::p_load(rio, here, dplyr, ggplot2, DT, tidyverse, formatR, gridExtra,
               pandoc, skimr, knitr, pandoc, flexdashboard, gt, flextable,
               reactable, kableExtra, janitor, ggpubr)
```

```
cleaned_NHANES <- import(here("data", "cleaned_NHANES.csv"))
cleaned_NHANES_avrg_SBP <- import(here("data", "cleaned_NHANES_avrg_SBP.csv"))
```

Creating Complex Tables

Overall sample size of each wave

```
# sample size of each wave

table_size_wave <- table(cleaned_NHANES$wave_id)
table_size_wave
```

```
##
## 2015 2016 2017
## 10175 9971 9254
```

```
# format table
table_size_wave <- kable(
  table_size_wave,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Wave ID", "Count"),
  caption = "NHANES sample size of each wave"
)

# style table

table_size_wave <- table_size_wave %>%
  kable_styling(
    latex_options = "hold_position",
    full_width = FALSE
  )
table_size_wave
```

Table 1: NHANES sample size of each wave

Wave ID	Count
2015	10175
2016	9971
2017	9254

Number and percentage of males

```
# sample size of each wave by gender
```

```
count_wave_gender <- table(cleaned_NHANES$wave_id, cleaned_NHANES$gender)
count_wave_gender
```

```
##
##      Female Male
##  2015   5172 5003
##  2016   5079 4892
##  2017   4697 4557
```

```
addmargins(count_wave_gender, margin = 2)
```

```
##
##      Female Male Sum
##  2015   5172 5003 10175
##  2016   5079 4892 9971
##  2017   4697 4557 9254
```

```
# row percentages
```

```
row_percent_gender <- prop.table(count_wave_gender, margin = 1) * 100
row_percent_gender
```

```
##
##      Female      Male
##  2015 50.83047 49.16953
##  2016 50.93772 49.06228
##  2017 50.75643 49.24357
```

```
#combine and only keep male stats
```

```
male_percent_size <- as.data.frame(cbind(
  n = rowSums(count_wave_gender),
  Male = round(row_percent_gender[, 2], 1)
))
male_percent_size
```

```
##      n Male
## 2015 10175 49.2
## 2016  9971 49.1
## 2017  9254 49.2
```

```

# Format table

male_percent_size <- kable(
  male_percent_size,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Count", "Percent Male"),
  caption = "Percentage of male participants by wave year"
)

# style table

male_percent_size <- male_percent_size %>%
  kable_styling(
    latex_options = "hold_position",
    full_width = FALSE
  )
male_percent_size

```

Table 2: Percentage of male participants by wave year

	Count	Percent Male
2015	10175	49.2
2016	9971	49.1
2017	9254	49.2

Number and percentage of each ethnicity for each wave

```

# sample size of each wave by ethnicity

count_wave_ethn <- table(cleaned_NHANES$wave_id, cleaned_NHANES$ethnicity_2)
count_wave_ethn

```

```

##
##      Asian Black Mexican Other Other Hispanic White
## 2015  1074  2267   1730   470           960  3674
## 2016  1042  2129   1921   505          1308  3066
## 2017  1168  2115   1367   634           820  3150

```

```

addmargins(count_wave_ethn, margin = 2)

```

```
##
```

```
##           Asian Black Mexican Other Other Hispanic White    Sum
##  2015  1074  2267    1730   470             960  3674 10175
##  2016  1042  2129    1921   505             1308  3066  9971
##  2017  1168  2115    1367   634             820  3150  9254
```

```
# row percentages
```

```
row_percent_ethn <- prop.table(count_wave_ethn, margin = 1) * 100
row_percent_ethn
```

```
##
##           Asian      Black    Mexican      Other Other Hispanic      White
##  2015 10.555283 22.280098 17.002457  4.619165      9.434889 36.108108
##  2016 10.450306 21.351921 19.265871  5.064688     13.118042 30.749173
##  2017 12.621569 22.854982 14.771990  6.851091      8.861033 34.039334
```

```
#combine
```

```
ethn_percent_size <- as.data.frame(cbind(
  n = rowSums(count_wave_ethn),
  Asian = round(row_percent_ethn[, 1], 1),
  Black = round(row_percent_ethn[, 2], 1),
  Mexican = round(row_percent_ethn[, 3], 1),
  Other = round(row_percent_ethn[, 4], 1),
  "Other Hispanic" = round(row_percent_ethn[, 5], 1),
  White = round(row_percent_ethn[, 6], 1)
))
```

```
ethn_percent_size
```

```
##           n Asian Black Mexican Other Other Hispanic White
## 2015 10175  10.6  22.3    17.0   4.6             9.4  36.1
## 2016  9971  10.5  21.4    19.3   5.1             13.1  30.7
## 2017  9254  12.6  22.9    14.8   6.9             8.9  34.0
```

```
# Format table
```

```
ethn_percent_size <- kable(
  ethn_percent_size,
  format = "latex",
  booktabs = TRUE,
  col.names = c(
    "Count",
    "Asian",
    "Black",
    "Mexican",
```

```

    "Other",
    "Other Hispanic",
    "White"),
  caption = "Count and percentage of each ethnicity for each wave"
)

ethn_percent_size <- ethn_percent_size %>%
  kable_styling(
    latex_options = "hold_position",
    full_width = FALSE
  ) %>%
  add_header_above(
    c(" " = 1, " " = 1, "Ethnicity" = 6)
  )

ethn_percent_size

```

Table 3: Count and percentage of each ethnicity for each wave

	Count	Ethnicity					
		Asian	Black	Mexican	Other	Other Hispanic	White
2015	10175	10.6	22.3	17.0	4.6	9.4	36.1
2016	9971	10.5	21.4	19.3	5.1	13.1	30.7
2017	9254	12.6	22.9	14.8	6.9	8.9	34.0

Summary statistics of average systolic blood pressure, by wave

```

#generate summary

summary_stats <- cleaned_NHANES_avrg_SBP %>%
  group_by(wave_id) %>%
  summarise(
    min = round(min(SBP_avg, na.rm = TRUE), 1),
    max = round(max(SBP_avg, na.rm = TRUE), 1),
    median = round(median(SBP_avg, na.rm = TRUE), 1),
    mean = round(mean(SBP_avg, na.rm = TRUE), 1),
    sd = round(sd(SBP_avg, na.rm = TRUE), 1)
  )

summary_stats

## # A tibble: 3 x 6
##   wave_id   min   max median   mean    sd

```

```
##      <int> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1    2015  64.7  229.   115.   118.   18.1
## 2    2016   74   231.   117.   120.   18.4
## 3    2017  72.7  234    118    122.   20.3

# format Table

summary_stats <- kable(
  summary_stats,
  format = "latex",
  booktabs = TRUE,
  col.names = c(
    "wave ID",
    "minimum",
    "maximum",
    "median",
    "mean",
    "SD"),
  caption = "Summary statistics of average systolic blood pressure, by wave"
)

summary_stats <- summary_stats %>%
  kable_styling(
    latex_options = "hold_position",
    full_width = FALSE
  ) %>%
  add_header_above(
    c(" " = 1, "Summary Statistic" = 5)
  )

summary_stats
```

Table 4: Summary statistics of average systolic blood pressure, by wave

wave ID	Summary Statistic				
	minimum	maximum	median	mean	SD
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

Bibliography

Health Statistics, National Center for. 2026. “NHANES Questionnaires, Datasets, and Related Documentation.” <https://wwwn.cdc.gov/nchs/nhanes/>.